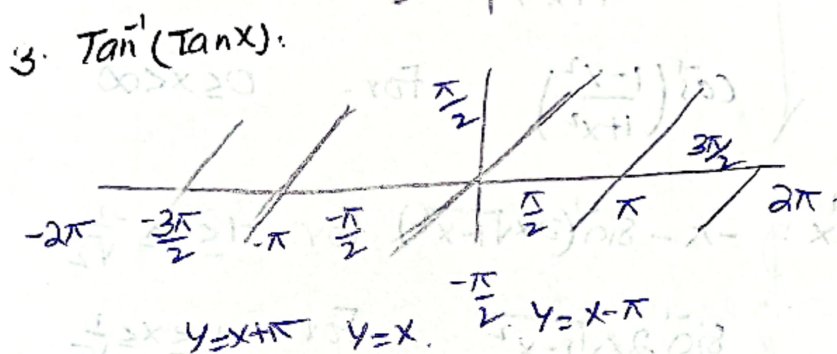
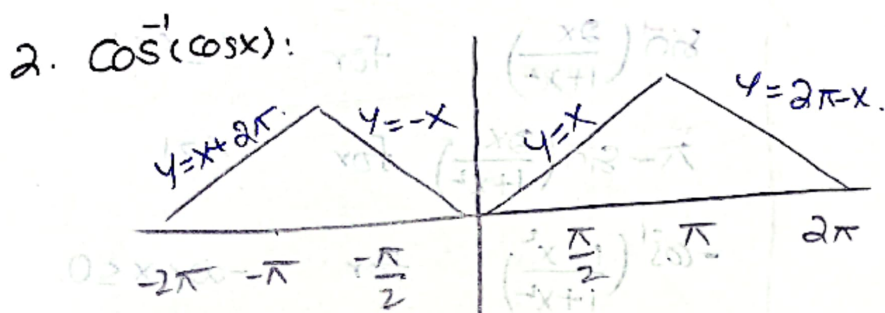
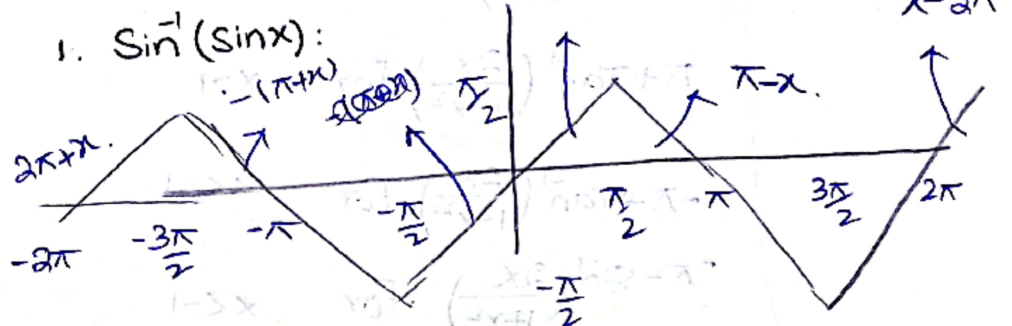


LESSON-18

Inverse Trigonometric Function

Graphs:



Properties:

$$x \in [-1, 1] \quad (i) \quad \sin^{-1} x + \cos^{-1} x = \frac{\pi}{2}$$

$$x \in \mathbb{R} \quad (ii) \quad \sec^{-1} x + \operatorname{cosec}^{-1} x = \frac{\pi}{2}$$

$$x \in (-\infty, -1) \cup (1, \infty) \quad (iii) \quad \cot^{-1} x + \tan^{-1} x = \frac{\pi}{2}$$

$$1. \quad \frac{\pi^3}{32} \leq (\sin^{-1} x)^3 + (\cos^{-1} x)^3 \leq \frac{7\pi^3}{8}$$

$$2. \quad \frac{\pi^2}{8} \leq (\sin^{-1} x)^2 + (\cos^{-1} x)^2 \leq \frac{5\pi^2}{4}$$

$$3. \quad -\frac{\pi^2}{4} \leq (\cos^{-1} x)^2 - (\sin^{-1} x)^2 \leq \frac{3\pi^2}{4}$$

$$1. \quad \sin^{-1}(-x) = -\sin^{-1} x$$

$$2. \quad \tan^{-1}(-x) = -\tan^{-1} x$$

$$3. \quad \operatorname{cosec}^{-1}(-x) = -\operatorname{cosec}^{-1} x$$

$$4. \quad \cos^{-1}(-x) = \pi - \cos^{-1} x$$

$$5. \quad \sec^{-1}(-x) = \pi - \sec^{-1} x$$

$$6. \quad \cot^{-1}(-x) = \pi - \cot^{-1} x$$

$$7. \quad \sin^{-1}\left(\frac{1}{x}\right) = \operatorname{cosec}^{-1} x$$

$$8. \quad \cos^{-1}\left(\frac{1}{x}\right) = \sec^{-1} x$$

$$9. \quad \tan^{-1}\left(\frac{1}{x}\right) = \cot^{-1} x$$

* * *

Idea:

$\sin^{-1}(\text{Anything})$

↓

Put Anything as Known Constant

Say,

Anything = 1

$$\cos^{-1} \frac{1}{2} = \cos^{-1} \frac{1}{2} = \frac{\pi}{3}$$

$$\cos^{-1} \frac{1}{2} + \cos^{-1} \frac{1}{2} = \frac{\pi}{3} + \frac{\pi}{3} = \frac{2\pi}{3}$$

$$\tan^{-1}x + \tan^{-1}y + \tan^{-1}z = \tan^{-1}\left(\frac{x+y+z-xyz}{1-xy-yz-zx}\right)$$

$$4. \ 2\tan^{-1}x = \begin{cases} \tan^{-1}\left(\frac{2x}{1-x^2}\right) & \text{for } -1 < x < 1 \\ \pi + \tan^{-1}\left(\frac{2x}{1-x^2}\right) & \text{for } x > 1 \\ -\pi + \tan^{-1}\left(\frac{2x}{1-x^2}\right) & \text{for } x < -1 \\ -\pi - \tan^{-1}\left(\frac{2x}{1-x^2}\right) & \text{for } x < -1 \\ \sin^{-1}\left(\frac{2x}{1+x^2}\right) & \text{for } -1 \leq x \leq 1 \\ \pi - \sin^{-1}\left(\frac{2x}{1+x^2}\right) & \text{for } x > 1 \\ -\cos^{-1}\left(\frac{1-x^2}{1+x^2}\right) & \text{for } -\infty < x \leq 0 \\ \cos^{-1}\left(\frac{1-x^2}{1+x^2}\right) & \text{for } 0 \leq x < \infty \end{cases}$$

$$5. \ 2\sin^{-1}x = \begin{cases} -\pi - \sin^{-1}(2x\sqrt{1-x^2}) & \text{for } -1 \leq x \leq -\frac{1}{\sqrt{2}} \\ \sin^{-1}(2x\sqrt{1-x^2}) & \text{for } -\frac{1}{\sqrt{2}} \leq x \leq \frac{1}{\sqrt{2}} \\ \pi - \sin^{-1}(2x\sqrt{1-x^2}) & \text{for } \frac{1}{\sqrt{2}} \leq x \leq 1 \end{cases}$$

$$6. \ 2\cos^{-1}x = \begin{cases} \cos^{-1}(2x^2-1) & \text{for } 0 \leq x < 1 \\ 2\pi - \cos^{-1}(2x^2-1) & \text{for } -1 \leq x < 0 \end{cases}$$

$$\begin{aligned} * \quad \tan^{-1}\left(\frac{1}{1+x(x+1)}\right) + \tan^{-1}\left(\frac{1}{1+(x+1)(x+2)}\right) + \dots + \tan^{-1}\left(\frac{1}{1+(x+n-1)(x+n)}\right) &= \tan^{-1}(x+n) - \tan^{-1}x \end{aligned}$$

- NOTE : For NO. of Solutions Get Graph of LHS Part then RHS Part and Get NO. of Point of intersections
For NO. of solutions Get Relations Mathematically
Solve Get the interval. Divide each Cases.

Properties : (Frequently Asked 2024, 2025)

1. $\sin^{-1}x + \sin^{-1}y$:

(i) For $0 \leq x, y, x^2 + y^2 \leq 1$ or $-1 \leq x, y \leq 1, xy < 0$ or $x^2 + y^2 > 1$

$$= \sin^{-1}\{x\sqrt{1-y^2} + y\sqrt{1-x^2}\}$$

(ii) For $0 < x, y \leq 1$ and $x^2 + y^2 > 1$

$$= \pi - \sin^{-1}\{x\sqrt{1-y^2} + y\sqrt{1-x^2}\}$$

(iii) For $-1 \leq x, y < 0$ and $x^2 + y^2 > 1$

$$= -\pi - \sin^{-1}\{x\sqrt{1-y^2} + y\sqrt{1-x^2}\}$$

2. $\sin^{-1}x - \sin^{-1}y$:

$$= \sin^{-1}\{x\sqrt{1-y^2} - y\sqrt{1-x^2}\} \quad 0 \leq x, y < 1 \text{ and } x^2 + y^2 > 1$$

$$= \pi - \sin^{-1}\{x\sqrt{1-y^2} - y\sqrt{1-x^2}\} \quad 0 < x \leq 1, -1 \leq y < 1, x^2 + y^2 > 1$$

$$= -\pi - \sin^{-1}\{x\sqrt{1-y^2} - y\sqrt{1-x^2}\} \quad 0 \leq y \leq 1, -1 \leq x < 0, x^2 + y^2 > 1$$

3. $\cos^{-1}x + \cos^{-1}y$:

$$: \cos^{-1}\{xy - \sqrt{1-x^2}\sqrt{1-y^2}\} \quad -1 \leq x, y \leq 1, x+y \geq 0$$

$$: 2\pi - \cos^{-1}\{xy - \sqrt{1-x^2}\sqrt{1-y^2}\} \quad -1 \leq x, y \leq 1, x+y < 0$$

4. $\cos^{-1}x - \cos^{-1}y$:

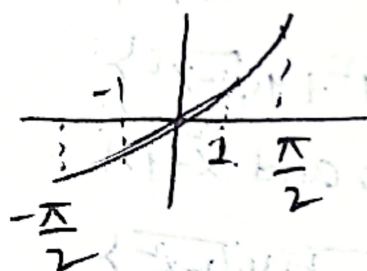
$$\cos^{-1}\{xy + \sqrt{1-x^2}\sqrt{1-y^2}\} \quad -1 \leq x, y \leq 1, x \leq y$$

$$- \cos^{-1}\{xy + \sqrt{1-x^2}\sqrt{1-y^2}\} \quad -1 \leq y \leq 0, 0 \leq x \leq 1, x \geq y$$

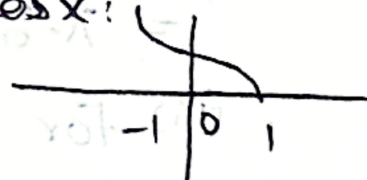
Function	Domain	Range
$\sin x$	$[-1, 1]$	$[-\frac{\pi}{2}, \frac{\pi}{2}]$
$\cos x$	$[-1, 1]$	$[0, \pi]$
$\tan x$	$\mathbb{R}$	$(-\frac{\pi}{2}, \frac{\pi}{2})$
$\operatorname{cosec} x$	$\mathbb{R}$	$(0, \pi)$
$\sec x$	$(-\infty, -1] \cup [1, \infty)$	$[0, \frac{\pi}{2}) \cup (\frac{\pi}{2}, \pi)$
$\cot x$	$(-\infty, -1] \cup [1, \infty)$	$[-\frac{\pi}{2}, 0) \cup (0, \frac{\pi}{2}]$

To figure out No. of Solution by Graph:

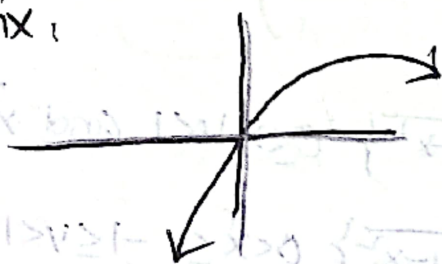
1. $\sin x$



2. $\cos x$



3. $\tan x$



4. $\cot x$

